Agentic AI Essentials

Demo-1

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Demo: Creating Python virtual environment on VS Code

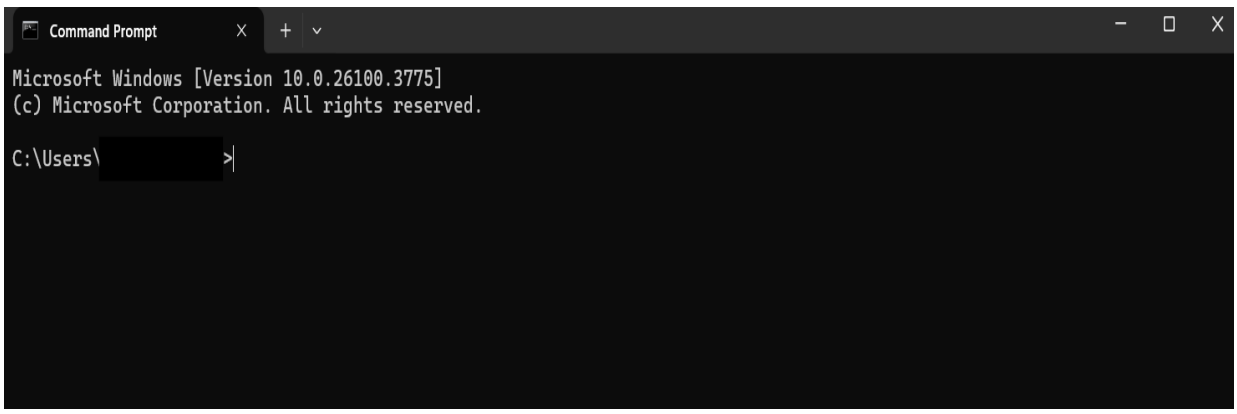
Why Use a Virtual Environment?

- Averts clashes among various project dependencies. Every project may utilize its unique collection of packages without affecting other projects or the global Python installation on the system.
- Make certain that every required package and its respective versions are documented in the environment folder as it makes it easier to share and replicate projects, as all dependencies are clearly defined.

Step-by-step guide for creating and virtual environment on VS Code:

Step 1: Open Command Prompt (CMD)

- Start with a clean Windows Command Prompt.
- The CMD is used for setting up project folders and creating the environment.



Step 2: Navigate to the Desired Drive

- Change the current drive to 'D:' using the command `D:`

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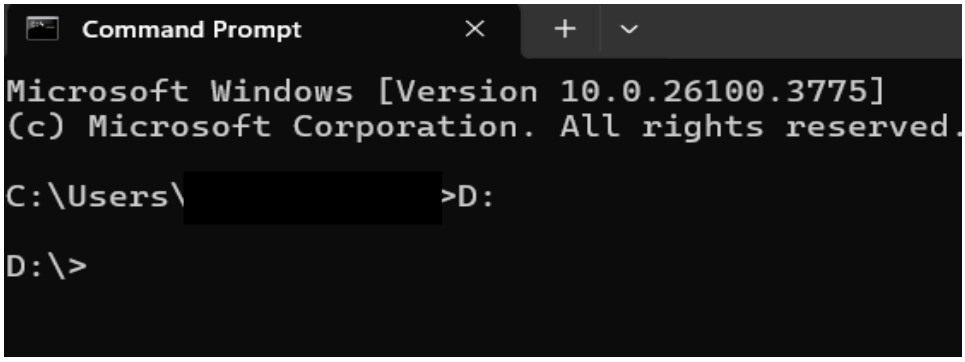
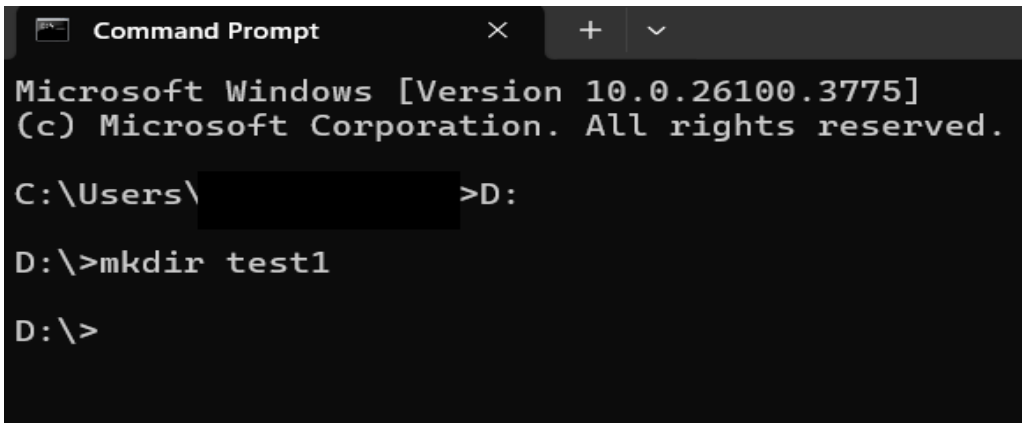
- You can use the drive you want to create your folder or virtual environment in.



Step 3: Create a project folder named "test1" using the command `mkdir test1`.



Step 4: Enter the project folder using the command `cd test1`.

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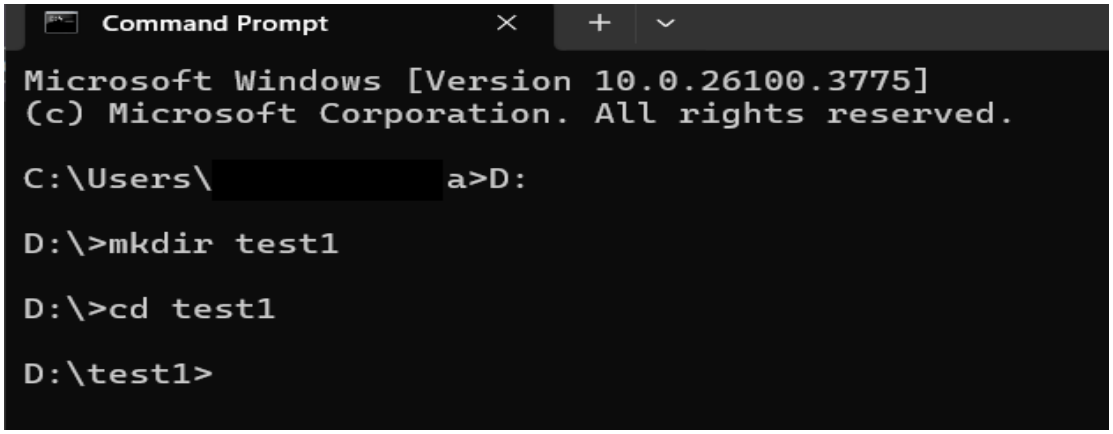
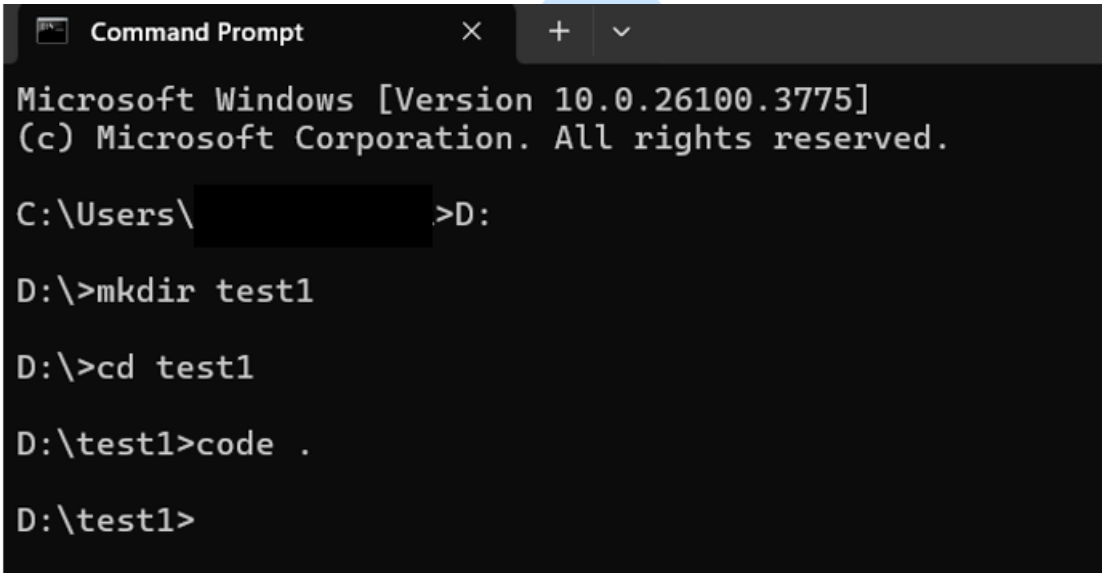


Step 5:

- Launch VS Code from the "test1" directory using the command '**code .**'
- This sets "test1" as the root directory and all future files are stored.
- You can do the same by directly creating folder from VS Code interface.



Step 6: VS Code Interface

Check the Explorer in VS Code has the "test1" as the folder name to verify that VS Code recognizes "test1" as the project and working directory.

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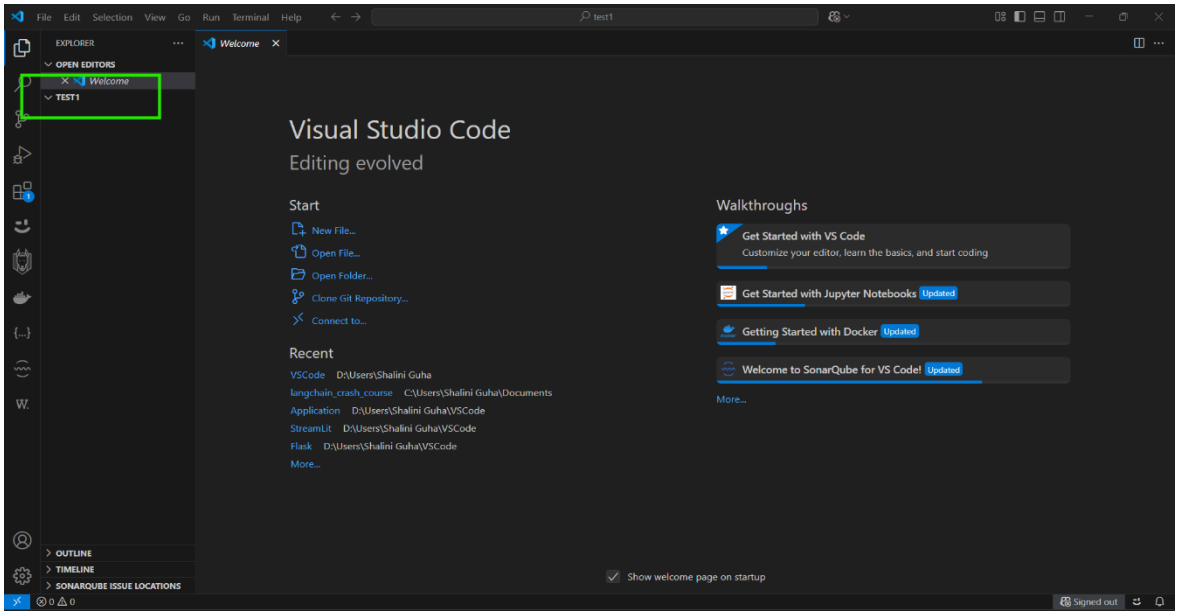
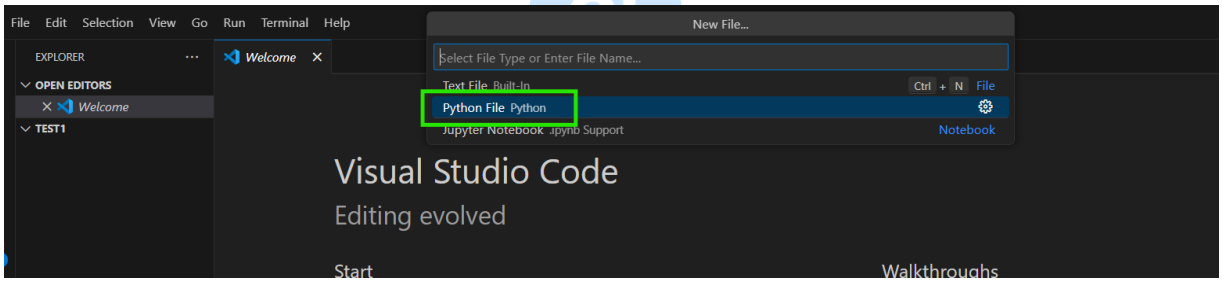
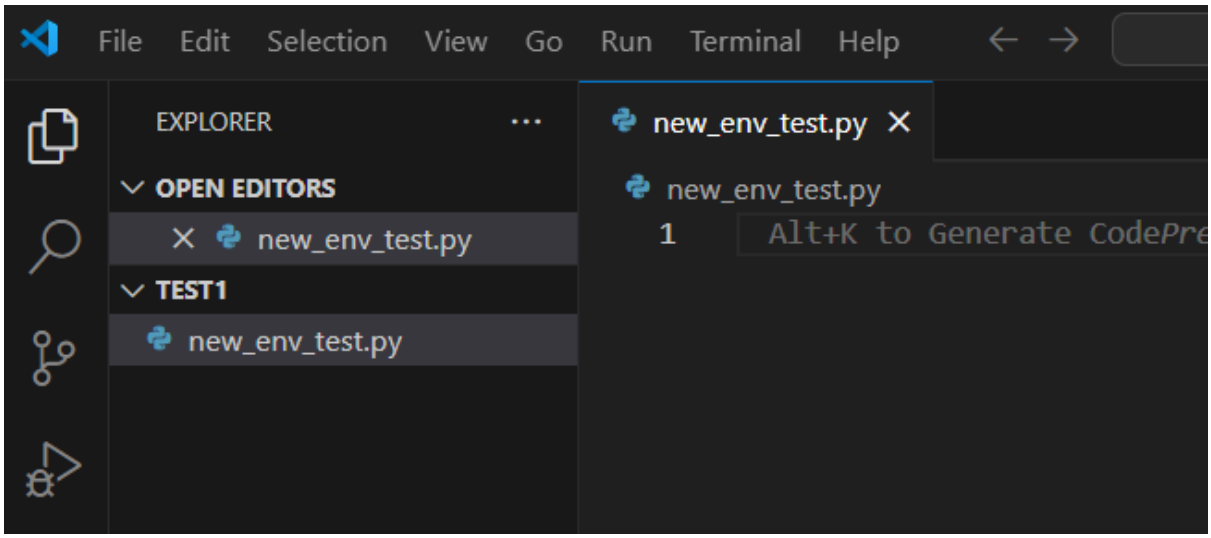


Step 7: Create Python file

In VS code create a new Python file (e.g. `new_env_test.py`)





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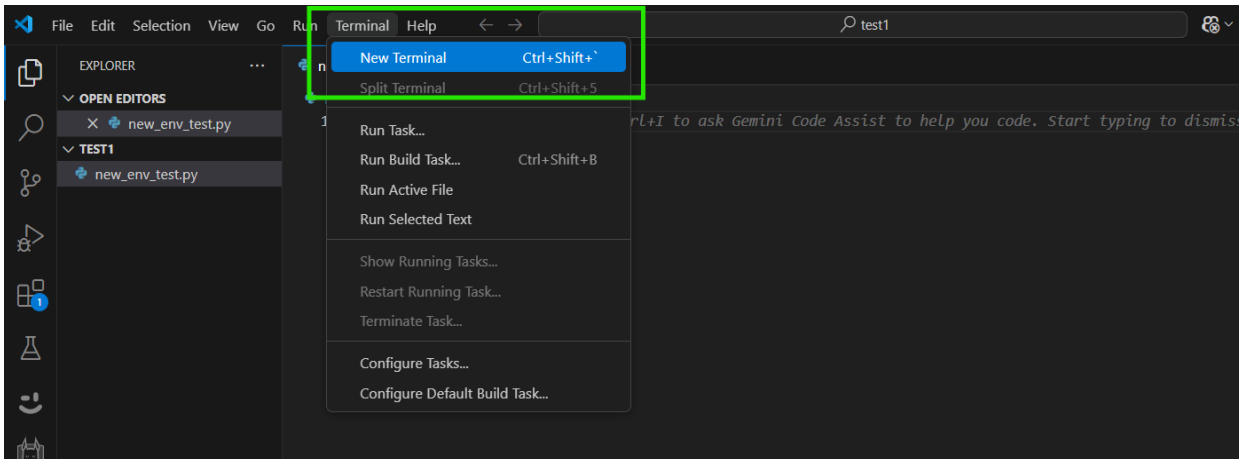
Step 8: Open Command Prompt Terminal

- Open a new command prompt (cmd) terminal in VS Code.
- VS code automatically opens powershell so you have to click on cmd and it is critical to create and activate in CMD.



Command Prompt Terminal Ready:

The Windows Command Prompt (cmd) is now open within VS Code's terminal panel, with the current directory set to `D:\test1` which is the selected environment for running the virtual environment commands.

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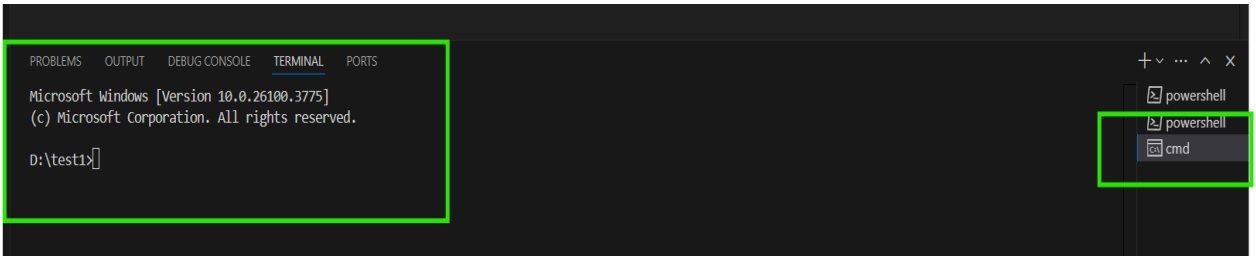

Step 9: Create Virtual Environment

- Execute the command python -m venv env1 within the VS Code terminal.
- This creates the "env1" directory containing the virtual environment's files.

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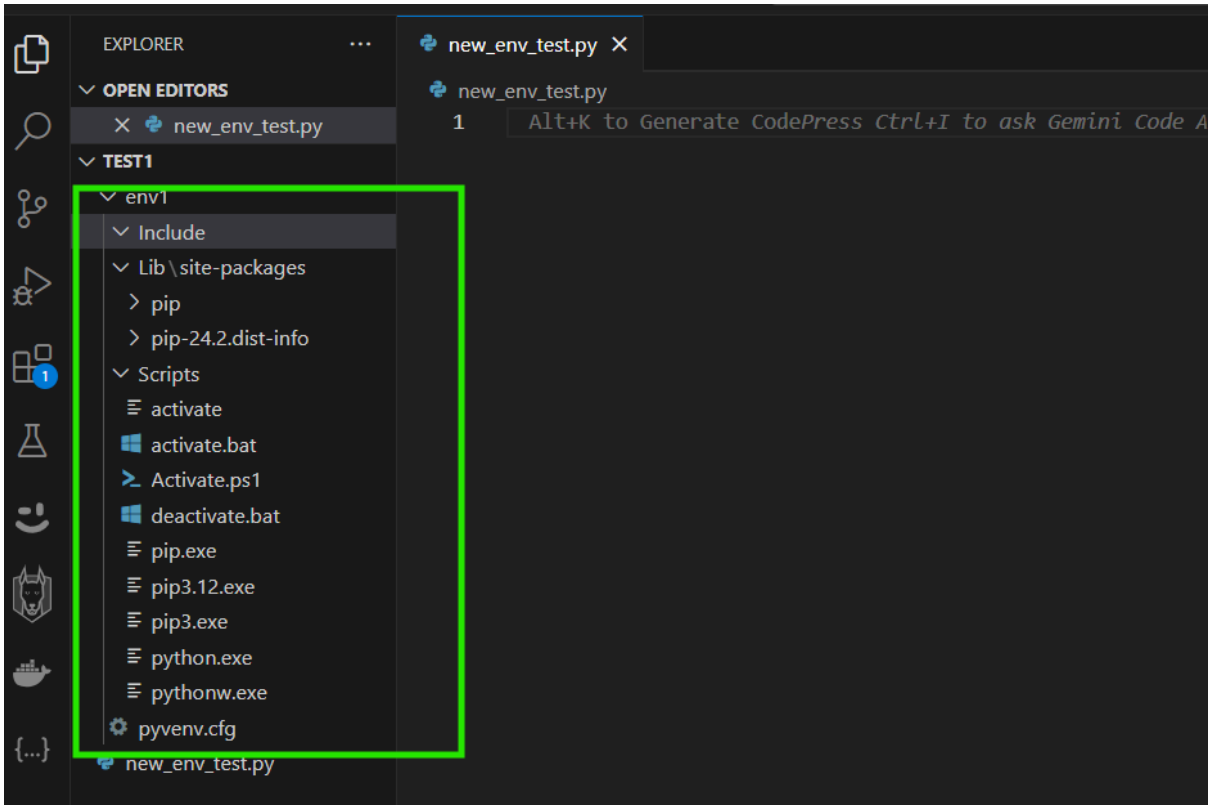

Step 10: Select Python Interpreter

- Open the Command Palette (`Ctrl+Shift+P`) and search for "`Python: Select Interpreter`."
- Select the Python interpreter located within your environment folder (e.g., "`.../env1/Scripts/python.exe`").
- This ensures that VS Code will run your Python code and manage packages within the environment.

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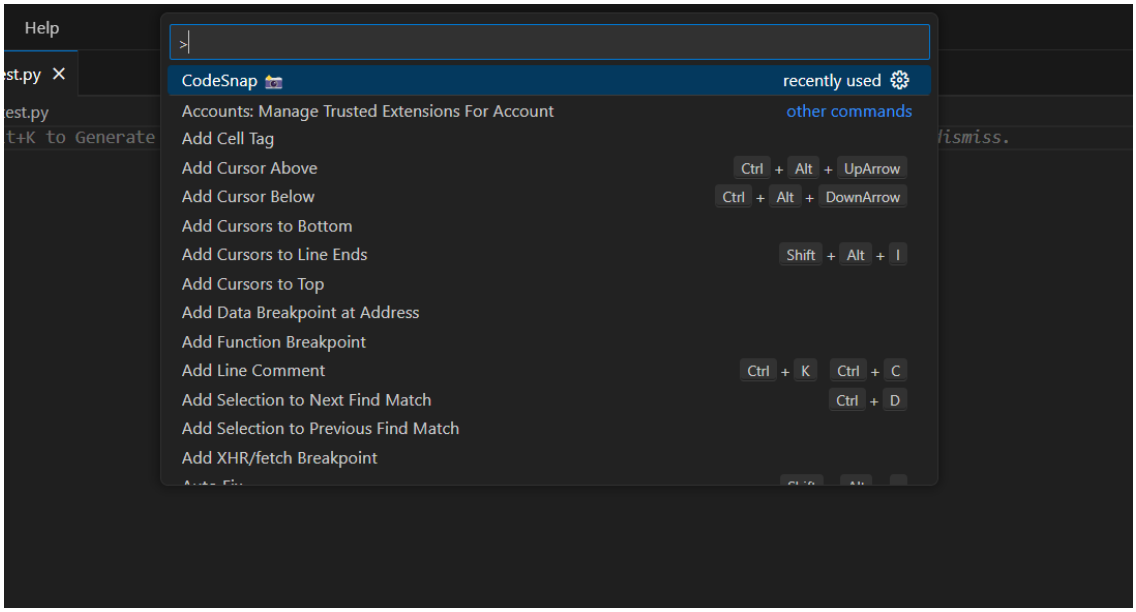

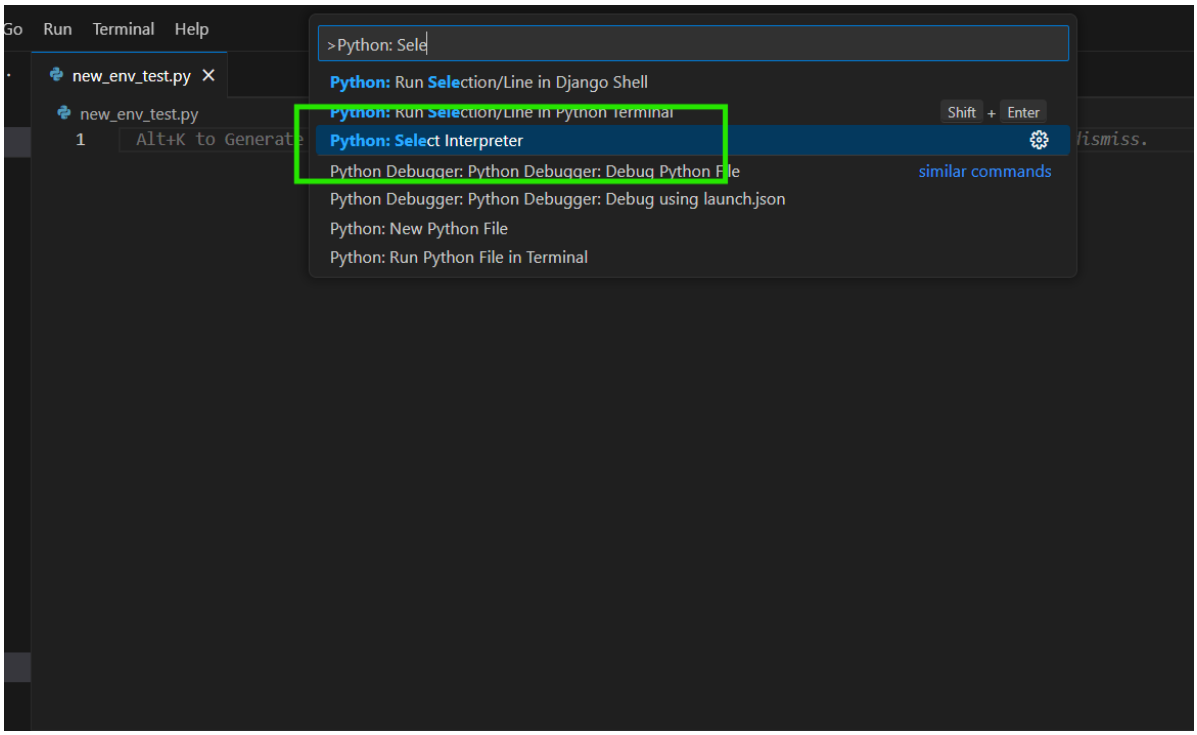

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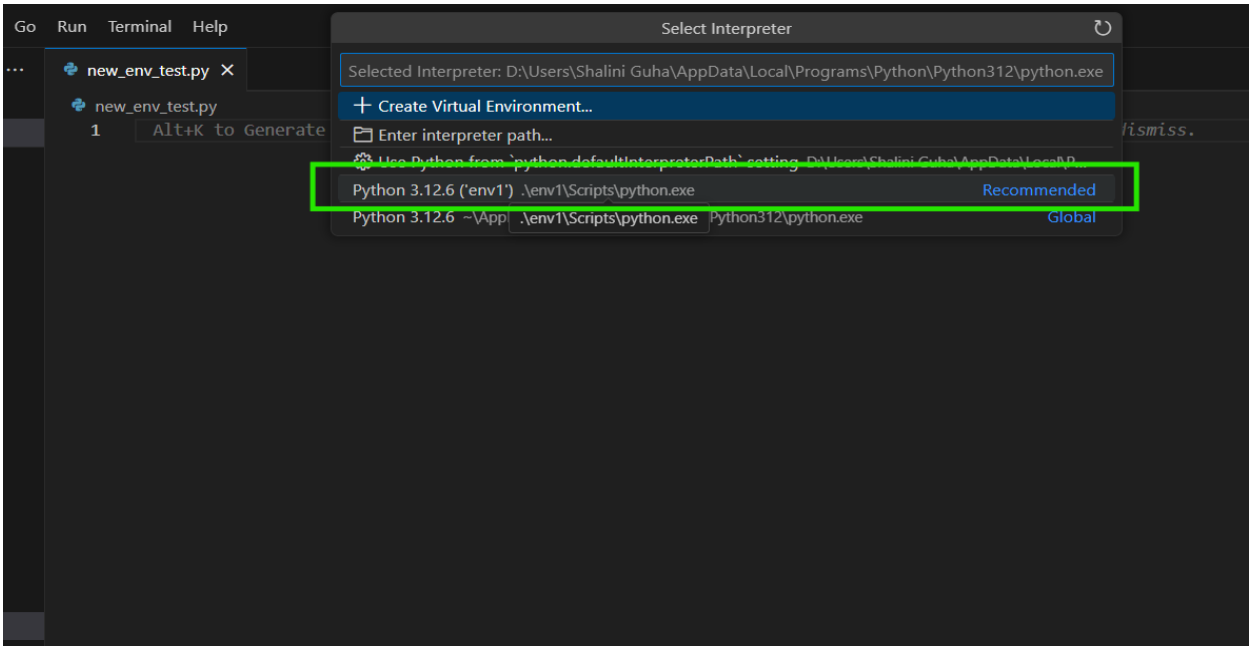

Step 11: Running code in Python (Powershell)

```
def add(x, y):
    return x + y

def subtract(x, y):
    return x - y

def multiply(x, y):
    return x * y

def divide(x, y):
    if y == 0:
        return "Error: Division by zero"
    return x / y

def main():
    print("Simple Calculator")
    print("Select operation:")
    print("1. Add")
    print("2. Subtract")
    print("3. Multiply")
    print("4. Divide")

    choice = input("Enter choice (1/2/3/4): ")

    if choice not in ['1', '2', '3', '4']:
        print("Invalid input")
```

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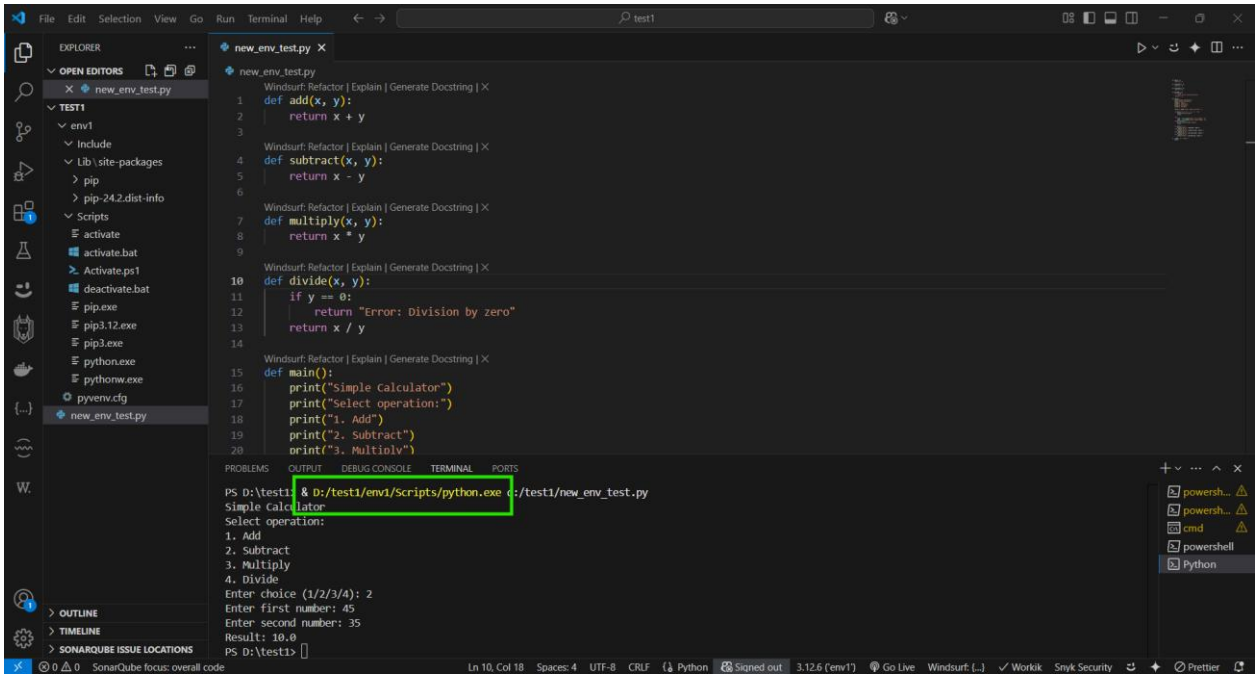
```
        return

    try:
        num1 = float(input("Enter first number: "))
        num2 = float(input("Enter second number: "))
    except ValueError:
        print("Invalid number input")
        return

    if choice == '1':
        print(f"Result: {add(num1, num2)}")
    elif choice == '2':
        print(f"Result: {subtract(num1, num2)}")
    elif choice == '3':
        print(f"Result: {multiply(num1, num2)}")
    elif choice == '4':
        print(f"Result: {divide(num1, num2)}")

if __name__ == "__main__":
    main()
```

- A python script has been run in powershell with the selected interpretor env1 in VS Code, and outputting the result in the VS Code Terminal Panel.
- It runs the script and executes the code and produces an output.



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Step 12: Activating the Virtual Environment

- In the cmd of VS Code, activate it by typing the command env1\Scripts\activate.
- Activating it will tell the system use the python interpretor that the virtual environment is being run from.

```
D:\test1>env1\Scripts\activate
```

After activating the environment with the `env1\Scripts\activate` command, the prompt changes to `(env1) D:\test1>`.

Difference in Prompt:

- `D:\test1>`: This is the standard command prompt, indicating the current directory (in this case, the "test1" project folder).
- `(env1) D:\test1>`: This indicates that the virtual environment "env1" is active. When the environment is active, pip installs are directed to the environment.

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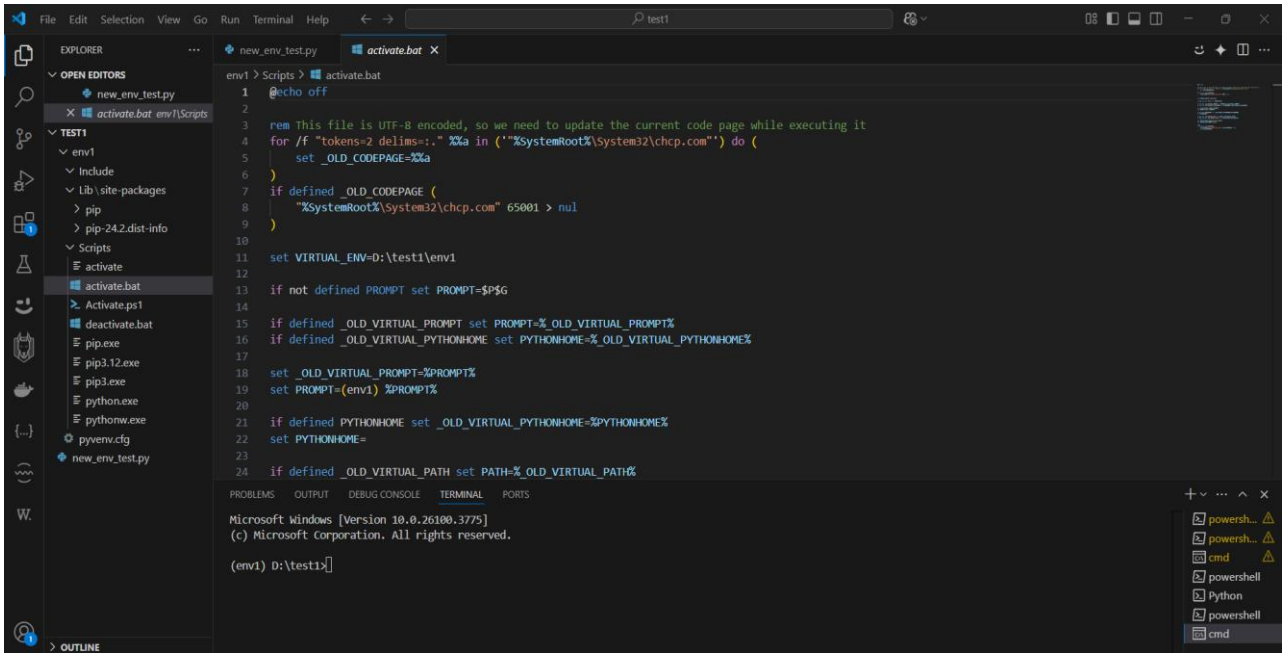
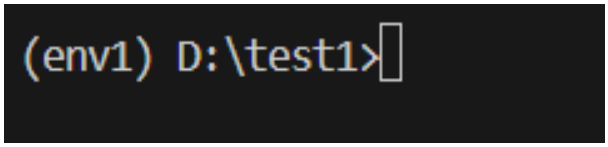
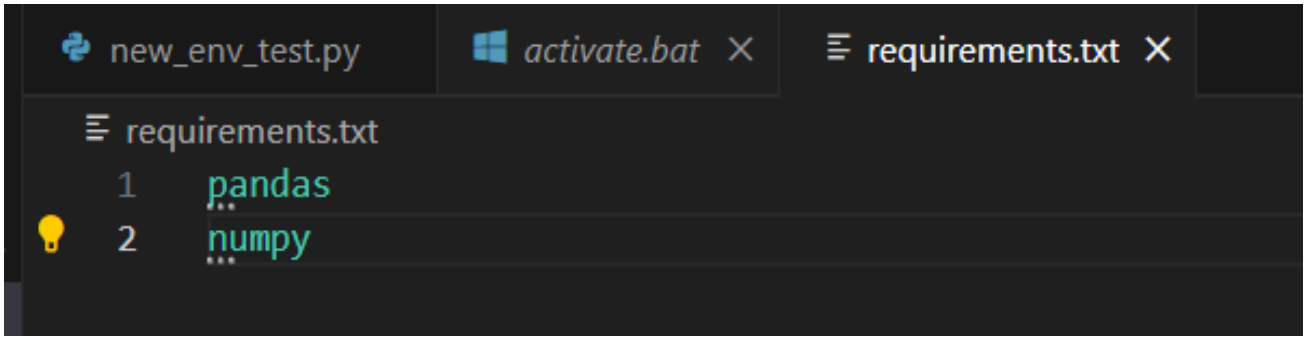




Step 13: Basic Requirements File

Creating a '`requirements.txt`' file for installations. For this demonstration, pandas and numpy is being used.




```
pandas
numpy
```

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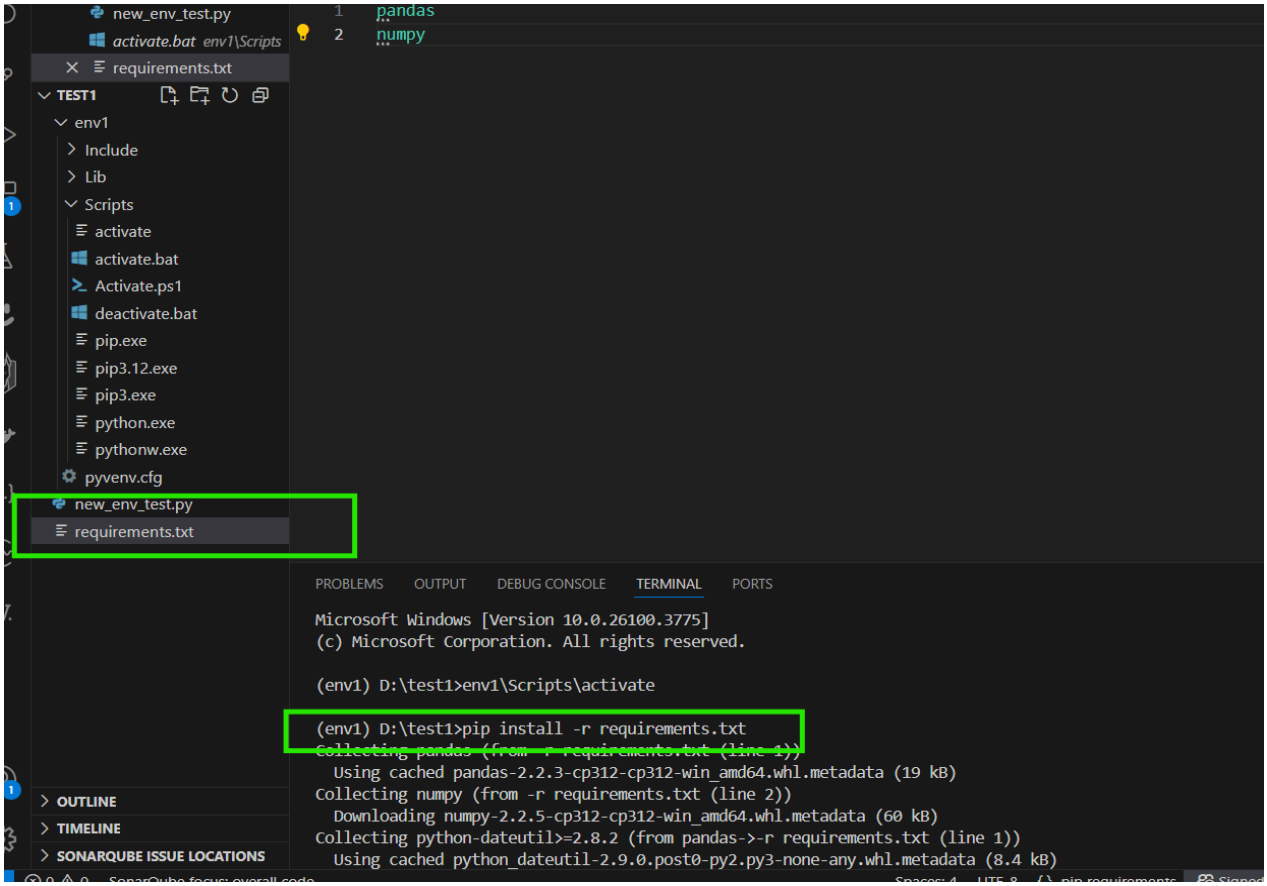
Step 14: Installing from requirements.txt

- Shows `pip install -r requirements.txt` command.
- This is an efficient way to install multiple required packages for the project into the activated environment.
- It creates the requirements.txt file to contain the packages to be downloaded, and executes the pip command to install.



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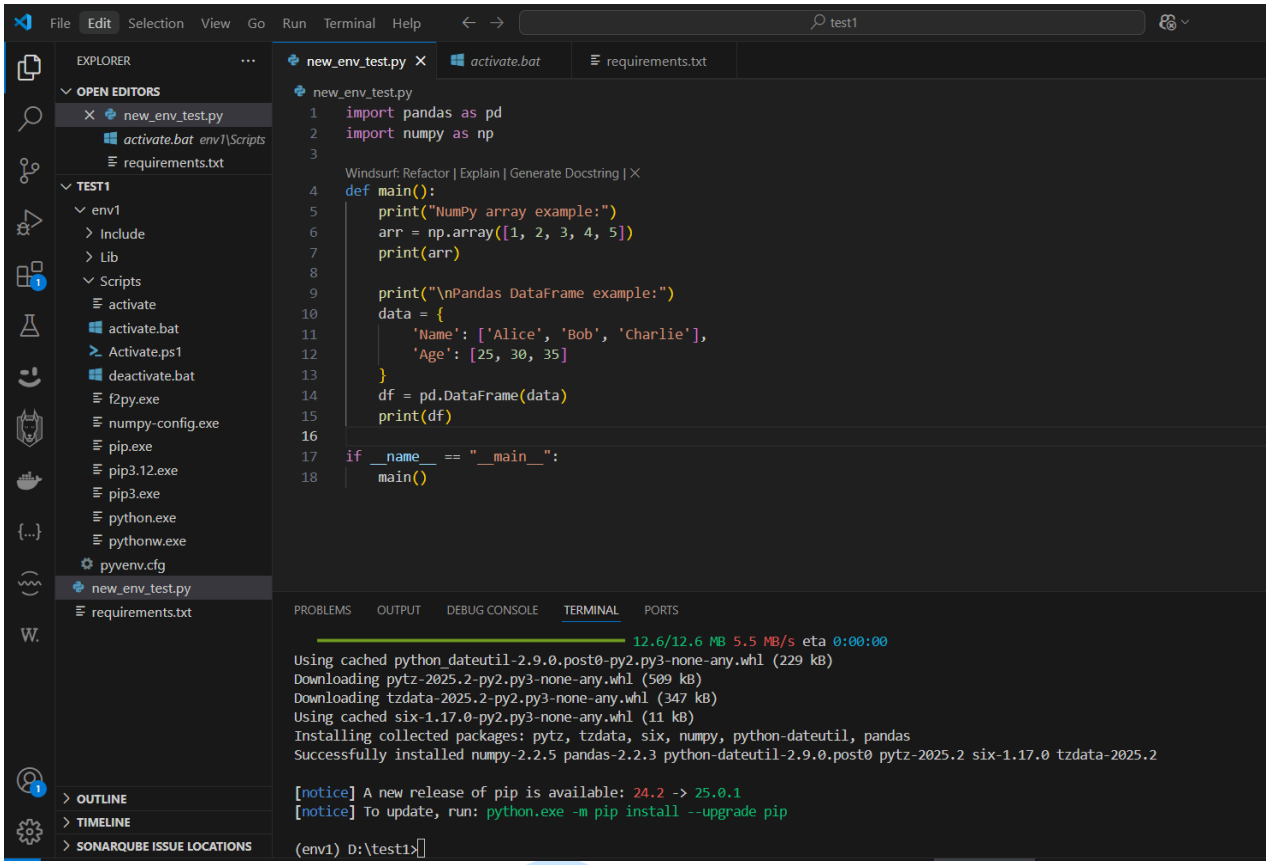

Step 15: Running a Python Script

```
import pandas as pd
import numpy as np

def main():
    print("NumPy array example:")
    arr = np.array([1, 2, 3, 4, 5])
    print(arr)

    print("\nPandas DataFrame example:")
    data = {
        'Name': ['Alice', 'Bob', 'Charlie'],
        'Age': [25, 30, 35]
    }
    df = pd.DataFrame(data)
    print(df)

if __name__ == "__main__":
    main()
```

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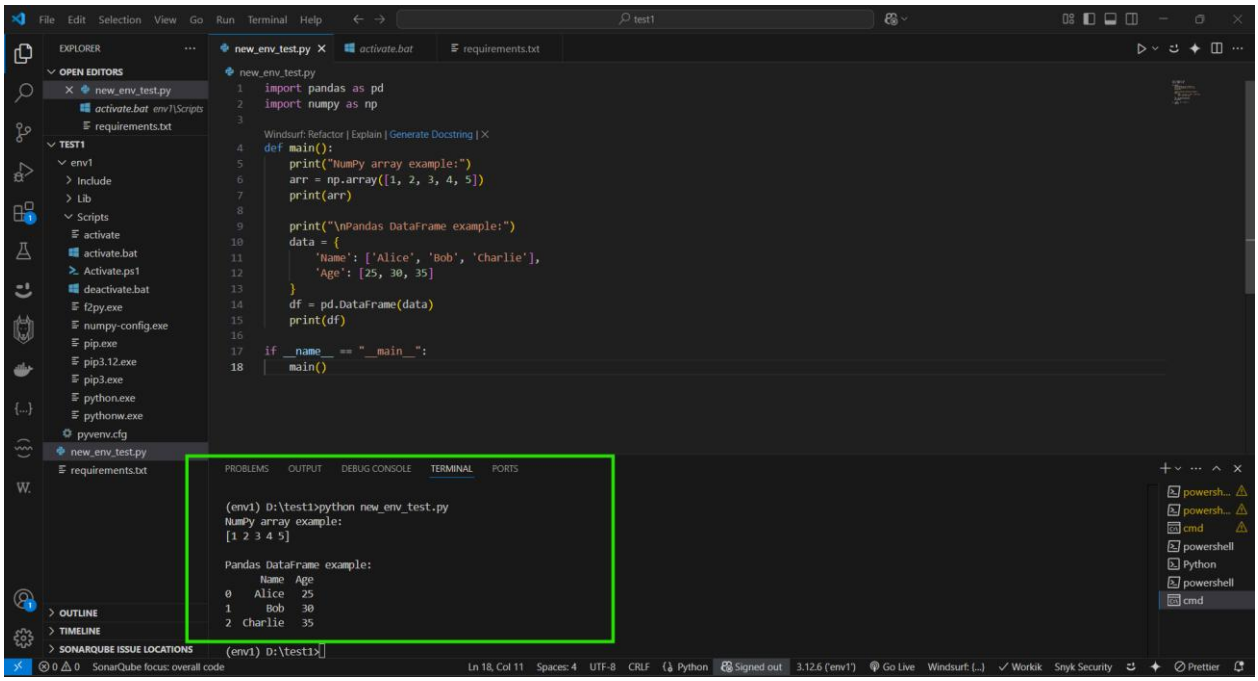
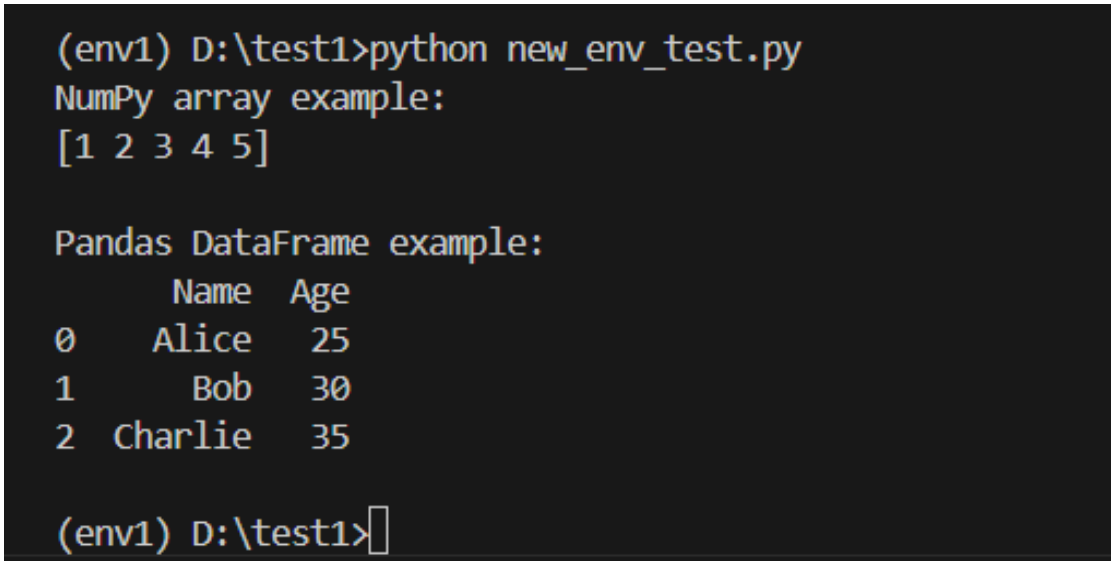
- Successfully running a Python script that imports and uses pandas and numpy, after they were installed into the environment.
- It validates correct installation and functionality of the packages within the activated environment and runs the Python script "new_env_test.py" in the powershell of VS Code which in turn selects the correct python interpreter and is able to run numpy and pandas.





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